

Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories — spending patterns, customer segments, and subscription behavior to guide strategic decisions.



Dataset at a Glance



3,900 Rows

Transactional records across 18 columns

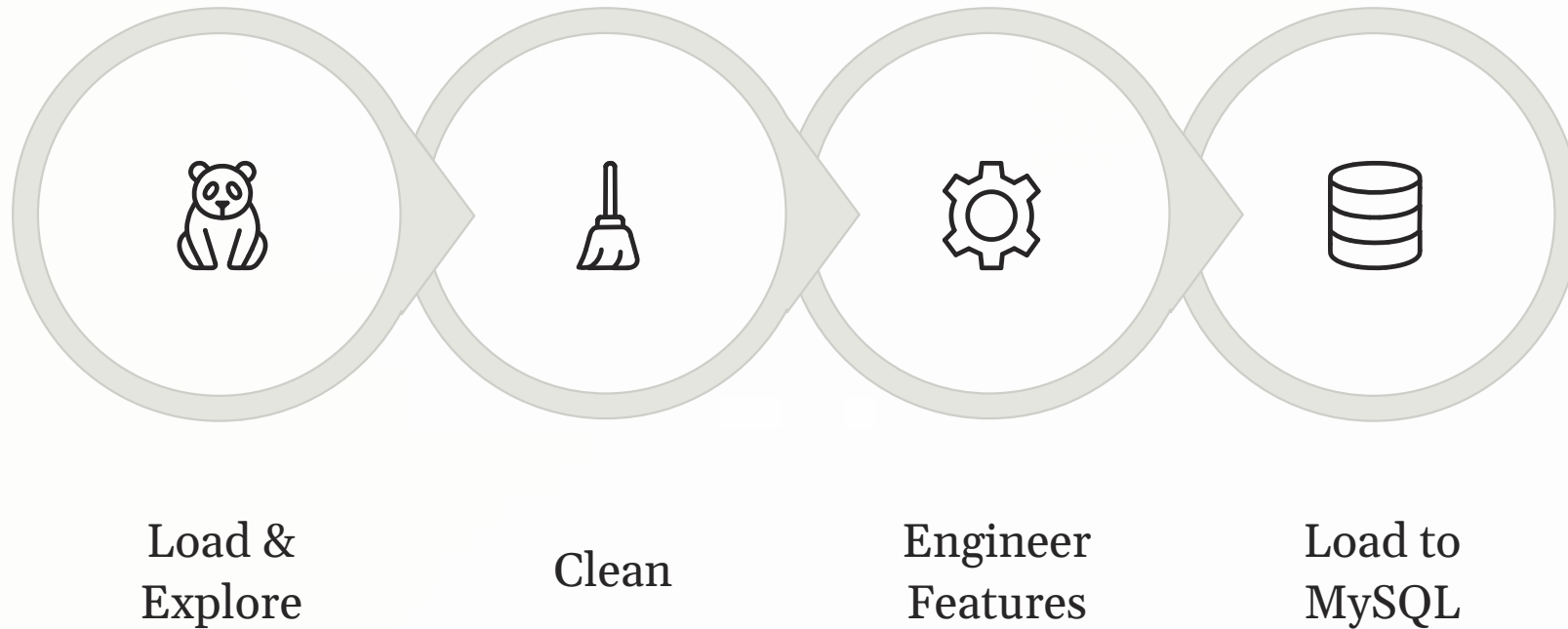
Key Features

Demographics, purchase details, shopping behavior, subscription status

37 Missing Values

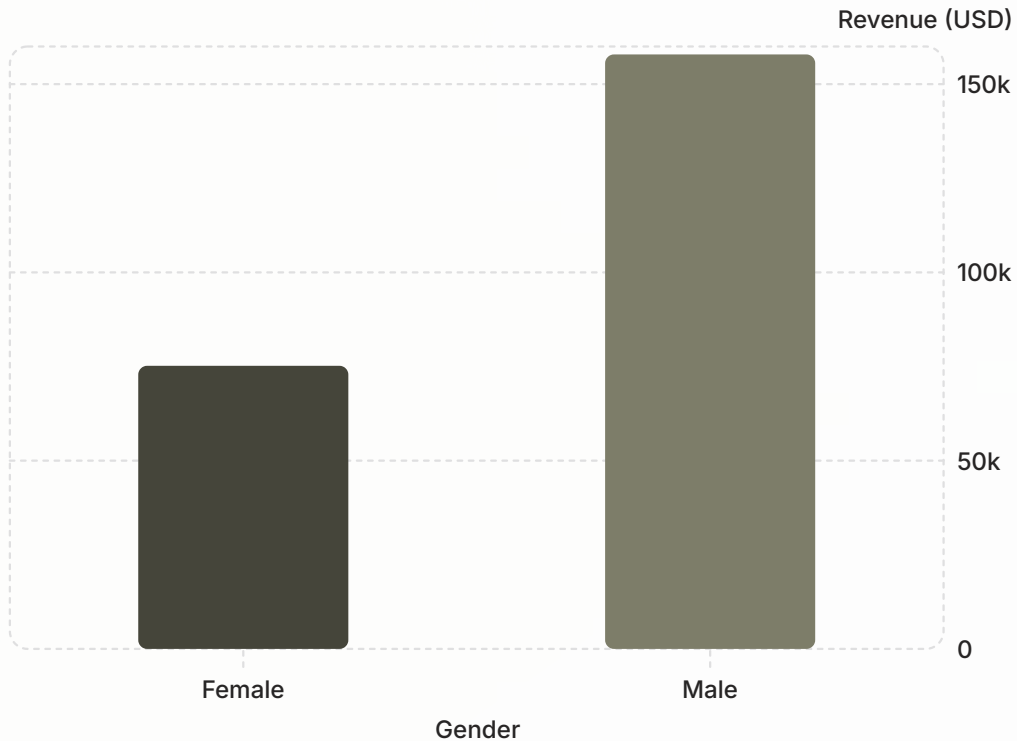
In Review Rating — imputed using median by product category

Data Preparation in Python



Columns renamed to snake_case; promo_code_used dropped as redundant with discount_applied. Cleaned DataFrame loaded into MySQL for SQL analysis.

Revenue by Gender

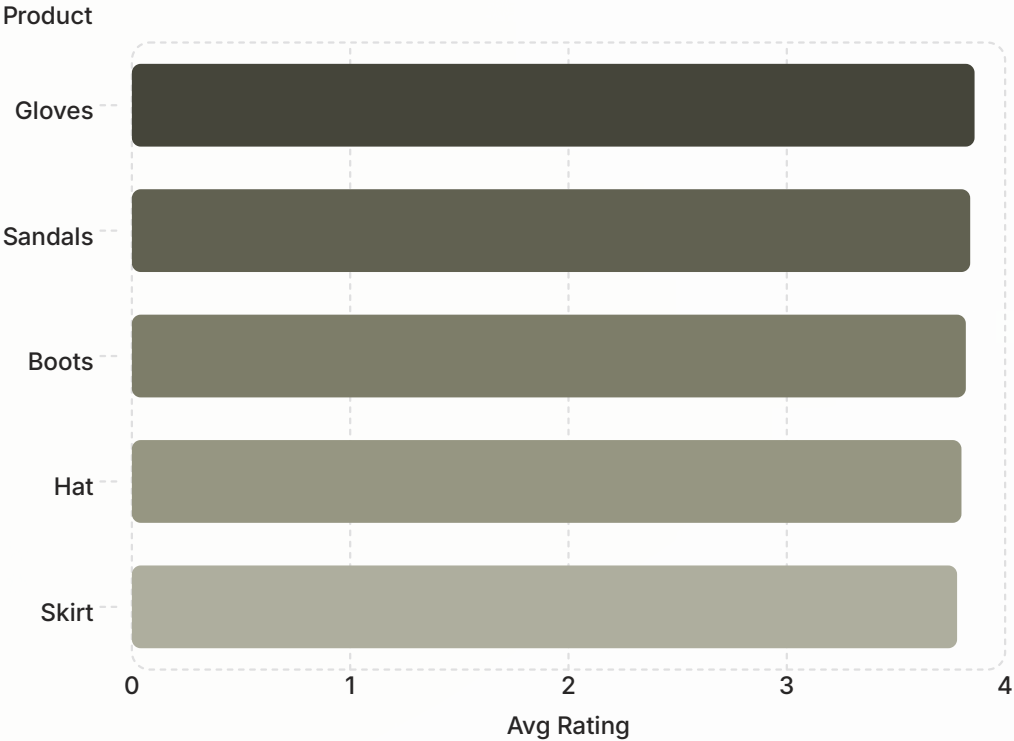


Key Finding

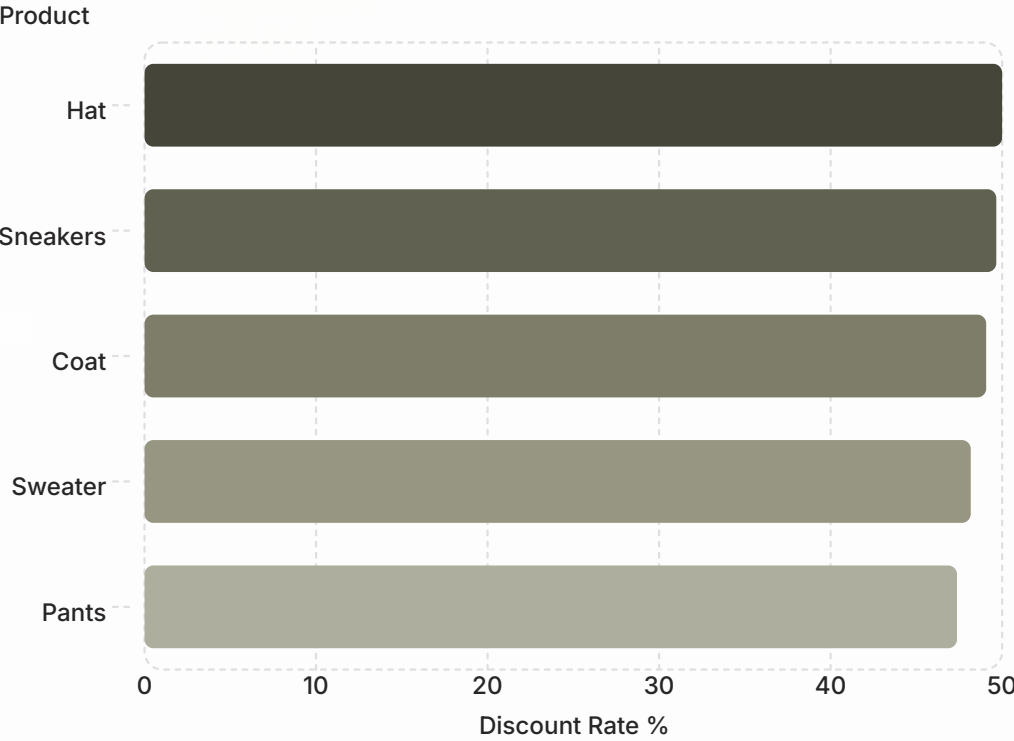
Male customers generated **more than double** the revenue of female customers – \$157,890 vs. \$75,191 – a critical insight for targeted marketing and product positioning strategies.

Top Products & Discount Patterns

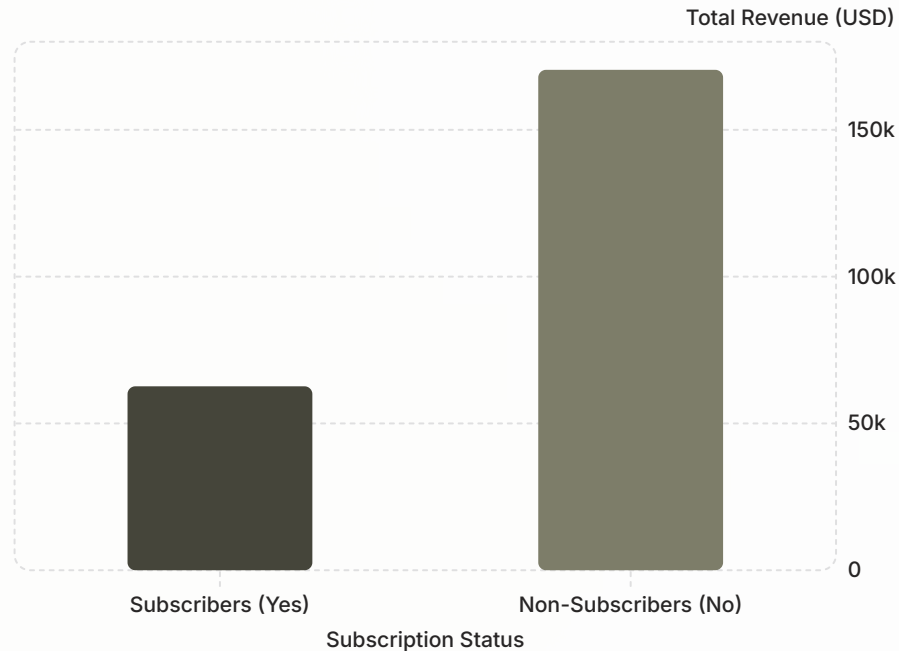
Top 5 Products by Average Rating



Highest Discount Rates



Subscribers vs. Non-Subscribers



Subscription Insights

Average spend is nearly identical across both groups, suggesting subscription benefits may not yet be compelling enough to drive conversion.

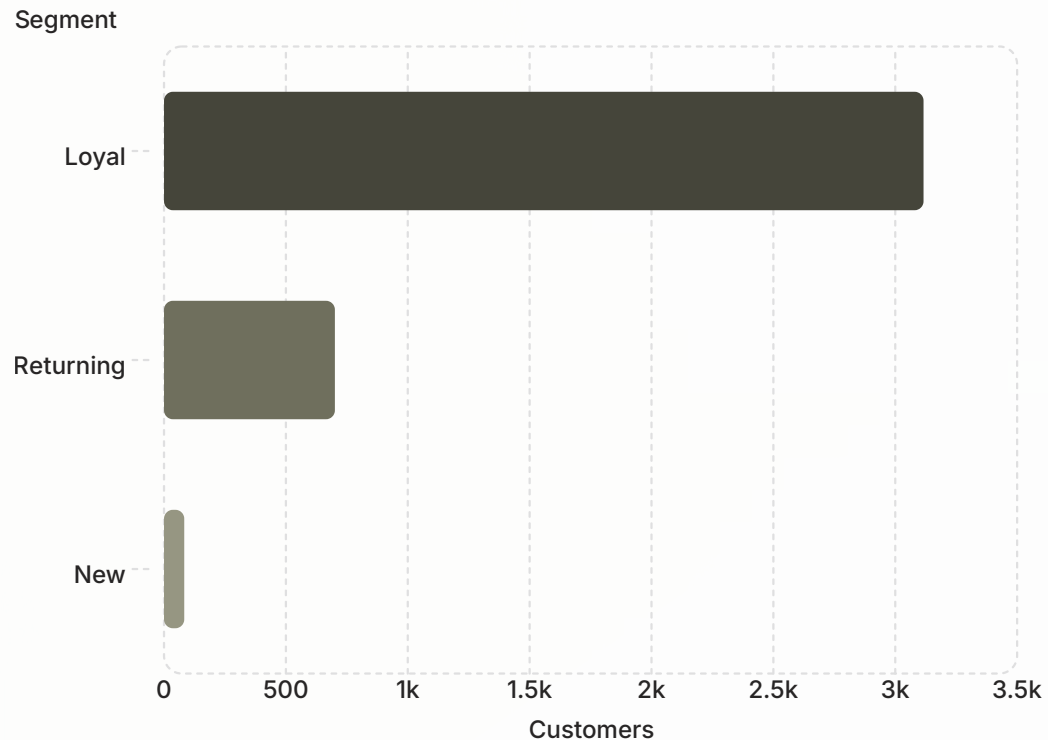
27% Subscribers

1,053 customers — avg spend \$59.49

73% Non-Subscribers

2,847 customers — avg spend \$59.87

Customer Segmentation

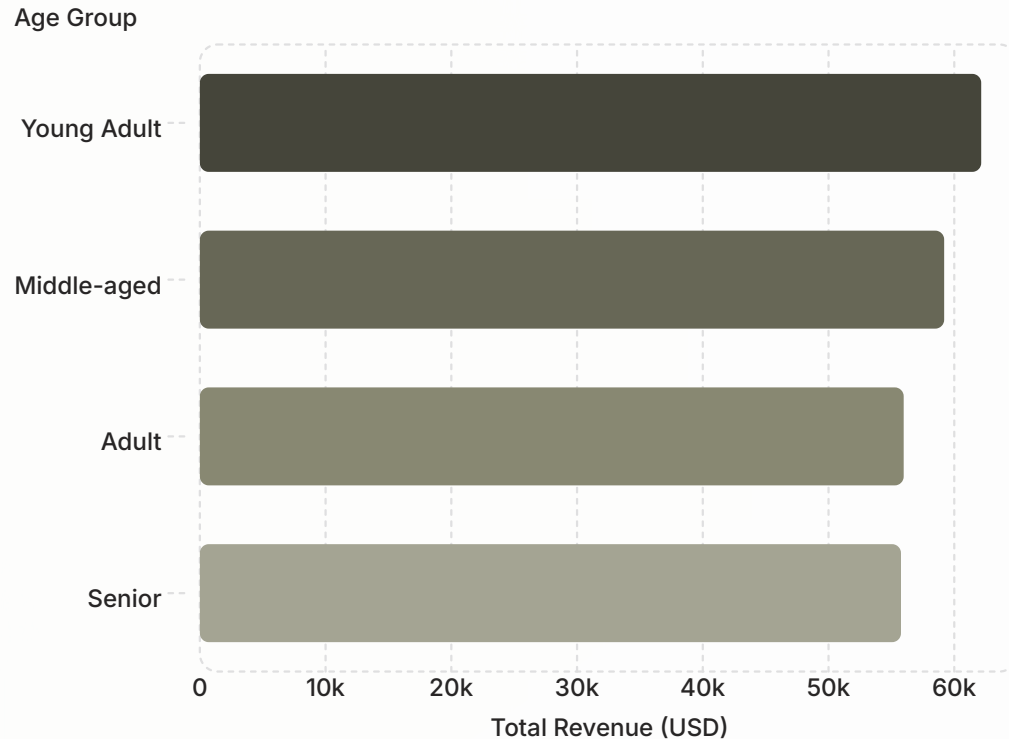


Segment Breakdown

- **Loyal (3,116)** – The dominant segment, representing the core customer base
- **Returning (701)** – Opportunity to convert into Loyal customers
- **New (83)** – Smallest segment; growth potential through acquisition

Of repeat buyers (>5 purchases), **958 are subscribers** vs. **2,518 non-subscribers** – a key conversion opportunity.

Revenue by Age Group & Shipping



Shipping Type Comparison

Express shipping customers spend slightly more on average (\$60.48) vs. Standard shipping (\$58.46).

- Young Adults lead revenue — prioritize in marketing
- Revenue is relatively evenly distributed across age groups
- Express shipping users represent a higher-value segment

Power BI Dashboard



Interactive Filters

Subscription status, gender, category, and shipping type



Key Metrics

3.9K customers • \$59.76 avg purchase • 3.75 avg rating



Visual Charts

Revenue & sales by category and age group; subscription donut chart



Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base.

Customer Loyalty Programs

Reward repeat buyers to move Returning customers into the Loyal segment.

Review Discount Policy

Balance sales boosts with margin control — nearly 50% discount rates on top products.

Product Positioning

Highlight top-rated products (Gloves, Sandals, Boots) in marketing campaigns.

Targeted Marketing

Focus on high-revenue age groups (Young Adults) and express-shipping users.